# Prompting an LLM

## Teaching with AI (TAI)

Athina Minaidi[1], Mirte van der Eyden[2], Andrea López Incera[3]

[1] 2nd Lyceum of Kos

[2] Institut für Fachdidaktik, Universität Innsbruck
mirte.van-der-eyden@uibk.ac.at

[3] Departament de didàctica de la matemàtica i les ciències experimentals, Universitat Autònoma de Barcelona
andrea.lopez.incera@uab.cat

Project funded by Erasmus+ Programme of the European Union

Subject: Any subject for which you use an LLM, or specific subjects like technology.
Activity duration: 45 min

Steps:

1. Prompting for an explanation of a topic (5 min)
2. Tips for good prompts (10 min)
3. Practice with example tasks (30 min)

Learning goals:

- Learn how to prompt LLMs to get the results you want
- Practice prompting
- Understand how language affects AI responses
- Develop critical thinking when comparing answers

# Prompting an LLM

*Tips for good prompting and practice*

In this activity, you will learn some tips to prompt LLMs, and you will practice prompting.

No matter what you ask and how you ask about it, the LLM will always give an answer (correct or incorrect). But ultimately, YOU are responsible for the answer, since you can influence the answer by tweaking the prompt, asking for sources or checking the factual correctness. Be critical and always check for external sources of the facts that the LLM provides.

## Assignment 1: Asking for an explanation of a topic

### a) First, choose a topic from the list below.

- Explain gaming addiction.
- Explain Quantum Physics.
- How does AI work?
- Why do people get addicted to social media?
- Help me study.

### b) Second, prompt the LLM for an explanation of the topic you chose above.

**Question 1: Write here the prompt you wrote asking the LLM for the explanation of the topic.**

**Question 2: Was the result you got to your liking? If not, why not?**

**Question 3: Would that explanation be ok for any person/audience?**

## Assignment 2: Tips for prompting

Together with this document, this lesson has a double-side card with tips for prompting LLMs.

**a) Read the card carefully.**

## Assignment 3: Update your prompt

**Question 4: Write the updated version of the prompt you wrote in Question 1, now taking into account the prompting tips. Try to add goal, context, format, style, examples, etc.**

**Question 5: Were the answers different? If yes, which step made the biggest difference? What does this tell you about how AI works?**

## Assignment 4: Asking the same question...but in different ways

**Question 6: Create 3 different prompts for the same topic, adapting each of them to a different audience (8, 17, and 20 year-olds).**

First, choose a topic from the list below:

1. Explain Black Holes
2. How the Universe Was Created
3. Explain Quantum Physics
4. Explain AI

- Topic
- 8 year-old child
- 17 year-old student

- 20 year-old university student

**a) Use a LLM to test your prompts. It will generate three answers.**

**Question 7: Compare the answers and explain what changed and why.**

## Assignment 5: Same question, different language

**Question 8: Ask the SAME question in 4 different languages (English, Danish, Spanish, Greek) and compare the answers.**

First, choose a topic from the list below:

1. What is the purpose of the European Union?
2. How does the European Union affect young people?
3. What are the advantages and disadvantages of living in the EU?

- Topic

- First, write the (upgraded) prompt in English

Then, translate it to different languages:

- Danish

- Greek

- Spanish

**Question 9: Compare the four answers generated by the AI in the four languages. Is the answer in some language clearer than in others? Is there**

**a language that has missing ideas in the answer? Check in the table below which options apply for each language.**

| **Language** | **Clearer** | **More detailed** | **Missing ideas** | **Translation differences** |
|---|---|---|---|---|
| English | | | | |
| Danish | | | | |
| Spanish | | | | |
| Greek | | | | |

Additional Notes and Observations:

**Question 10: Quality gap: Which language provided the best overall answer quality? Why do you think one language was clearer or with better answer quality than another?**

**Question 11: (Extension) What happens if we write the prompt in English... but ask for the answer in another language? Does the quality improve compared to writing the prompt in Greek from the start?**